Documentation Report

Of Final Project/Exam:VB.NET Application Development for Sustainable Development

Submitted by:

Alba, Jimmy Paul

Doton, Athan Josch

Lizardo, Gerald

Navarro, Aaron Christian

Santos, Justine

Submitted to:

Ms. Justin Louise R. Neypes

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1.1 Project Overview & UN SDG Target

The Community Resource Management System is a Windows Forms application developed in VB.NET that leverages ADO.NET for persistent data handling and Report Viewer for professional reporting. Designed as part of the Capstone Synthesis for the VB.NET Application Development course, the system addresses operational challenges faced by local communities in managing sustainable initiatives.

Unlike manual record-keeping, which often results in inefficiencies and data inconsistencies, this application provides a structured, secure, and user-friendly platform for managing resources, beneficiaries, and distribution processes. By integrating object-oriented programming principles, modular architecture, and robust error handling, the system ensures reliability and maintainability in real-world deployment scenarios.

1.2 Problem Statement

Local government units and community organizations often encounter difficulties in managing sustainable resource programs:

1. **Manual Tracking Issues** – Paper-based records lead to errors, duplication, and loss of critical data.
2. **Inefficient Processing** – Verification and reporting consume significant time, delaying aid and resource distribution.
3. **Data Integrity Risks** – Lack of validation allows duplicate entries, incorrect records, and unreliable statistics.
4. **Unstructured Reporting** – Absence of automated reporting prevents stakeholders from making informed decisions.
5. **Resource Mismanagement** – Poor monitoring leads to shortages, surpluses, or inequitable allocation of resources.

Proposed Solution

The Community Resource Management System addresses these challenges through:

- **Automated CRUD Operations** – Secure database integration via ADO.NET with parameterized queries.
- **Business Logic Modules** – Dedicated classes for resource validation, beneficiary eligibility, and consumption tracking.
- **Queue and Priority Handling** – Structured distribution order ensuring fairness and prioritization of vulnerable groups.
- **Professional Reporting** – Microsoft Report Viewer integration for generating summary and trend reports.
- **Robust Error Handling** – Comprehensive Try-Catch-Finally blocks with clear, actionable error messages.
- **Data Persistence & Security** – Configuration-based connection strings and role-based login system for controlled access.

III. Objectives

3.1 General Objective

To develop a Community Resource Management System that empowers organizations with tools to manage resources efficiently, monitor consumption, and promote sustainable practices aligned with SDG 12.

3.2 Specific Objectives

1. Implement secure user authentication with role-based permissions.
2. Enable full CRUD operations for resource tracking via ADO.NET.
3. Provide a budget setup and monitoring dashboard with real-time feedback.
4. Integrate reporting through Microsoft Report Viewer for structured insights.
5. Generate smart allocation suggestions based on usage history.
6. Deliver category-based analysis and visual trend charts.

7. Support administrators with a centralized dashboard for oversight.
8. Enforce strict input validation and robust error handling.

IV. SDG Alignment

UN SDG Alignment: SDG 12 – Responsible Consumption and Production

This project directly contributes to UN Sustainable Development Goal 12: Responsible Consumption and Production, which emphasizes reducing waste, improving resource efficiency, and promoting sustainable practices. By enabling communities to track resource allocation, monitor consumption trends, and generate actionable reports, the system supports equitable distribution and minimizes mismanagement.

ID	**Requirements**	**Description**
FR1	**Data Persistence**	Full CRUD operations via ADO.NET with parameterized
FR2	**Business Logic**	Rule-based modules for allocation and monitoring.

V. Scope and Limitations

5.1 Scope

The system is a VB.NET Windows Forms application integrated with Microsoft SQL Server via ADO.NET, designed for community organizations and administrators. It supports resource tracking, budget monitoring, reporting, and role-based access.

5.2 Limitations

- No real-time API integration; data is manually entered.
- Desktop-only application; no web or mobile version.
- Rule-based suggestions only; no AI or machine learning.
- Limited to internal database; no external feeds.

VI. REQUIREMENT & ANALYSIS

6.1 Functional Requirements

ID	Requirements	Description
FR1	**Data Persistence**	Full CRUD operations via ADO.NET with parameterized
FR2	**Business Logic**	Rule-based modules for allocation and monitoring.
FR3	**Reporting**	Professional reports via Report Viewer.
FR4	**Security**	Role-based login system.
FR5	**Resource Tracking**	CRUD for resource entries.
FR6	**Budget Monitoring**	Real-time dashboard with indicators.
FR7	**Smart Suggestions**	Rule-based recommendations.
FR8	**Admin Dashboard**	Centralized oversight of system activity.
FR9	**Validation & Error Handling**	Strict input validation and Try-Catch-Finally

		blocks.

6.2 Non-Functional Requirements

ID	Requirements	Description
NFR1	**UI Consistency**	Standard naming conventions and clean layout.
NFR2	**Maintainability**	XML comments and modular code.
NFR3	**Reliability**	Config-based connection strings.
NFR4	**Performance**	Handle 100+ records without lag.
NFR5	**Usability**	Intuitive interface for non-technical users.
NFR6	**Security**	Parameterized queries and password hashing.

VII. Core Features

Feature	Description	Primary User
Login & User Management	Secure authentication and role-based access	Student, Admin
Resource Tracking	CRUD operations for resource entries	Student
Budget Setup	Configure daily/weekly/monthly budgets	Student

Budget Monitoring Dashboard	Real-time comparison of budget vs. usage	Student
Smart Suggestions	Rule-based recommendations	Student
Category Analysis	Charts showing spending/resource distribution	Student
Reporting System	Printable reports via Report Viewer	Student, Admin
Admin Dashboard	Oversight of users and resources	Admin

VIII. Tools and Technologies

Tool / Technology	**Category**	**Role in the System**
VB.NET	Programming Language	Core application logic and forms
Visual Studio	IDE	Development and debugging
Windows Forms	UI Framework	Graphical interface
SQL Server	Database	Persistent storage
ADO.NET (SqlClient)	Data Access Layer	CRUD operations
Report Viewer	Reporting Tool	Professional reports
GitHub	Version Control	Repository and collaboration

IX. Database Design

9.1 Entity Relationship Diagram (ERD).

9.2 Data Dictionary (Table names, fields, types, and constraints).

X. System Architecture (Flowcharts showing the Data Flow (DFD Level 0 and Level 1).

XI. Individual Contributions